

Ticket Pulse Mail Workflows QA Note

Strict incident signal detector and unsupported broader-issue wording guard

Date	June 5, 2026	Area	Settings > Mail Workflows and workflow audit
Branch	codex/strict-broader-issue-signals	Scope	Detector replacement, output guard hardening, QA PDF

Purpose	Validate that routine repeated work is no longer presented as a possible broader issue, while true incident-like clusters still produce strict, auditable signal evidence. Friendly requester tone remains allowed; the tightened rule is factual wording only.
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What Changed

Area	Change	QA should verify
Detector logic	Old broad count/diversity scoring was replaced with strict incident-language matching.	Routine clusters should not become requester-facing broader-issue claims.
Routine clusters	Common operational patterns now classify as routine_cluster: docking stations, peripherals, procurement, access, software installs, printers, webcams, licenses, and similar work.	Ticket 225627-style docking/peripheral examples should show routine_cluster and no allowed public phrase.
True incident signal	possible_broader_issue now requires explicit incident language plus shared open related incident tickets, 3 distinct requesters, and 2 distinct departments.	Only service-down/outage/cannot-connect style clusters with enough evidence should pass.
Allowed public wording	The only allowed public phrase is exactly: we are seeing multiple similar reports.	Generated copy must not say broader issue, wider issue, widespread issue, confirmed outage, or company-wide outage.
Output guard	Unsupported broader/similar-report wording is repaired or blocked and tagged as similar_report_claim_without_evidence.	Audit should show the guard rule when the LLM tries unsupported wording.
Settings UI	Labels now show Incident signal checks and Routine cluster threshold instead of the old broader-issue/watch wording.	Admins should see the new labels under Mail Workflows > LLM Context.

Where To Check In The App

- Settings > Mail Workflows > LLM Context: confirm the evidence source is labeled Incident signal checks and the numeric setting is Routine cluster threshold.
- Workflow Audit: open an LLM generate step and inspect outageSignals.signalLevel, outageSignals.criteria, outageSignals.counts, incidentSignals, thresholds, and allowedPublicPhrases.
- Workflow Audit signal filter: possible_broader_issue is displayed to operators as Strict incident signal; routine_cluster remains Routine cluster.

- Workflow Health: the old broader-issue-rate health text should appear as Strict incident rate when the threshold is crossed.

QA Scenarios

Scenario	Setup or sample	Expected result
Routine hardware/peripheral cluster	Use ticket 225627 / TP-NWF-309 as the reference pattern, or test a docking station, printer, webcam, or workstation setup request with several similar tickets.	signalLevel is routine_cluster; allowedPublicPhrases is empty; email does not mention broader issue, wider issue, widespread issue, or similar reports.
Routine access/software/procurement cluster	Use a normal access request, license request, software install, quote, invoice, or onboarding/procurement ticket with several similar matches.	signalLevel is routine_cluster or none; no requester-facing similar-report phrase is allowed.
Weak incident cluster	Use an incident-like ticket such as VPN cannot connect, but with one requester or one department only.	signalLevel is watch; allowedPublicPhrases is empty; requester copy speaks only about the current ticket.
True incident cluster	Use an outage/service-down/cannot-connect ticket with at least 3 open strong related tickets, 3 distinct requesters, and 2 distinct departments.	signalLevel is possible_broader_issue / Strict incident signal; allowedPublicPhrases contains only we are seeing multiple similar reports.
LLM attempts unsupported wording	In preview/manual test, use a prompt that nudges the model to say this may be part of a broader issue.	Guard repairs or blocks the sentence and audit records similar_report_claim_without_evidence with blockedBy/rule details.
Evidence source disabled	Turn off Incident signal checks for an LLM node or workspace policy and run preview.	No outage signal evidence should be supplied; LLM must not make any similar-report or outage claim.

Audit Fields To Confirm

- [] outageSignals.criteria.currentIncidentLanguage is true only when the current ticket has explicit incident wording such as outage, down, unavailable, cannot connect, or multiple users impacted.
- [] outageSignals.criteria.sharedIncidentSignal is true only when related tickets share the same incident signal family.
- [] outageSignals.counts.openIncidentStrongTickets is at least 3 before Strict incident signal is allowed.
- [] outageSignals.counts.distinctIncidentRequesters is at least 3 and distinctIncidentDepartments is at least 2 before public similar-report wording is allowed.
- [] outageSignals.allowedPublicPhrases is empty for routine_cluster, watch, none, or disabled evidence.
- [] When a repair happens, guard audit includes similar_report_claim_without_evidence and the final requester copy no longer contains the unsupported sentence.

Pass Criteria

- Routine operational clusters are never described to requesters as broader, wider, widespread, or similar-report issues.
- A true incident cluster can use only the exact deterministic phrase: we are seeing multiple similar reports.
- The QA team can explain each signal from the audit counts and criteria without reading backend logs.
- Friendly or relaxed tone still works; factual unsupported-impact claims remain protected.
- No regression in LLM generate fallback handling, SendGrid delivery, mock mode, routing, or workflow archive/delete behavior should appear during smoke tests.

Implementation Reference

- Backend detector: `backend/src/services/notificationContextEnrichmentService.js`
- Output guard: `backend/src/services/notificationWorkflowOutputGuard.js`
- Policy normalization and tool label: `backend/src/services/notificationLlmToolPolicyService.js`
- Health label/API message: `backend/src/routes/notificationWorkflow.routes.js`
- Settings and audit UI labels: `frontend/src/components/settings/NotificationWorkflowsPanel.jsx`
- Focused tests: `backend/tests/notificationContextEnrichmentService.test.js` and `backend/tests/notificationWorkflowOutputGuard.test.js`

**Known
compatibility
note**

The internal enum `possible_broader_issue` is intentionally retained for existing audit data and filters. Operators should see it labeled as Strict incident signal in the UI.